

Bruce Williams

CCIE #10701



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Summary

I offer over 20 years of experience in Network Engineering and Network Architecture. My background includes network design, network implementation, network management, project management, team management, pre-sales support, and post-sales support.

Certifications

CCIE Certification # 10701 – R&S / Enterprise Infrastructure, CCNA, CCDP, Cisco Certified Specialist - Data Center Core, Cisco IP Telephony, Cisco Certified Specialist - Enterprise Core, Wireless LAN for SEs and FEs

Accomplishments, Skills and Experience

- 20+ years experience Data Center Architecture, Design, Planning and Implementation
- 20+ years experience in network hardware; Cisco routers, switches, ASA/FTD, SDWAN, WLCs; Juniper routers, switches, firewalls; Palo Alto firewalls and Fortinet Firewalls.
- Served as Lead Architect and SME for MPLS, SDWAN.
- Designed and implemented migration from private cloud to public cloud. Migrated technologies and hardware included Nexus, VMWare, UCS and FPIC.
- Created detailed documentation including spreadsheets, diagrams, high level design and low level design documents.
- Led proof of concept testing and certification, providing certification testing plans, analysis and results documentation.
- Designed and implemented a service provider internal MPLS VPN network for a very large enterprise network. Configured all devices, P, PE, C and CE. Designed, planned the mutual redistribution between MBGP and OSPF. Configured RDs, RTs and planned import and export policies to provide redundancy and high availability.
- Designed and built a SIP based VoIP network for regional service provider.
- Analyzed and planned migrations of remote sites from Palo Alto to a Cisco SDWAN and Umbrella solution. Deployed new policies, templates and configs to Cisco VEdge and Cisco Umbrella. Established IPsec tunnels from VEdge devices to Umbrella. Managed VEdge devices using VManage.
- Designed and implemented Cloudgenix SDWAN solution, migrating network from point-to-point connections and MPLS to an overlay and underlay SDWAN network utilizing IPsec over Public Internet and Private MPLS VPN Network.
- Configured network connections and routing between customer premise and cloud provider both AWS and Azure. Configured AWS Gateway and Azure Express Route with BGP.
- Designed and implemented the migration from Cisco Catalyst 6509s to Nexus 7710s and FEX 2248s with multiple VDCs and VPCs.
- Architected, designed, and configured all devices for an IPv6, MPLS, OSPF/OSPFv3, BGP backbone for a large enterprise corporation.
- Analyzed, redesigned, and optimized a complex network of Cisco ASA Firewalls and Security Policies then migrated to Cisco Firepower for multiple large enterprise networks.

- Designed and deployed a Wireless LAN with a Cisco Wireless LAN Controller and Cisco Lightweight Access Points for a regional VoIP service provider.
- Designed, configured, and deployed a Juniper Dynamic Multipoint VPN for a large fabric manufacturer.
- Provided presales support and post-sales support to Cisco Resellers and Integrators in the areas of Routing, Switching, Data center, Security, Wireless and Unified Communications.
- Configured Authentication, Authorization and Accounting (AAA) on Cisco devices using Cisco ISE.
- Tested and certified network equipment which included Cisco WLAN Controllers, ISE, Cisco APs, Cisco Catalyst 9800 WLC Switches.
- Utilized Laptop Software with WiFi RSSI, and Spectrum Analyzers to measure and test RF levels for WiFi networks
- Extensively used packet captures, Ethereal, Wireshark, Cisco NAM, Cisco Netflow, Cisco NBAR and Cisco IP SLA to analyze the performance of applications through the network and make necessary changes to configurations and QoS Policies as necessary.

Job Experience

Telamon 03/25 – Present

Network Engineer Operations Team Lead

Remote

- Provided consulting services which consisted of design, configuration, implementation and installation of network equipment in AT&T Central Offices and Data Center.
- Configured Cisco, Juniper and Palo Alto network equipment.
- Supervised, Mentored and Trained engineers on the installation and configuration of network equipment; Cisco, Juniper and Palo Alto.
- Monitored network status with Solar Winds
- Troubleshoot network issues; analyze and resolve routing issues, BGP, IS-IS, OSPF; provide Root Cause Analysis.
- Created scripts with commands to automate the initial troubleshooting and display real time status of interfaces, routing tables, CPU, memory and logs.

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Hackensack Meridian Health Systems 12/24 – 02/25

Network Engineer /Architect

Hackensack, NJ

- Provided day to day support, troubleshooting, configuration of the Hackensack Health Systems network; Cisco routers and switches
- Configured and supported VPNs, SDWAN, MPLS, BGP.

Southwest Airlines / Oscar Technologies 3/24 – 11/24

Sr. Network Engineer

Remote

Hired as a contractor as lead engineer in the migration of Southwest Airlines, focusing on BGP redesign, migration of configurations of BGP, OSPF, and Catalyst Fabric.

- Utilized Cisco Nexus Dashboard Fabric Controller to deploy VXLAN/EVPN fabric, automate configuration and manage and monitor devices.
- Managed and monitored the network post migration to resolve issues.
- Designed, documented, implemented, and maintained complex routing and switching solutions.
- Served as the primary technical resource for BGP and OSPF routing protocols.
- Provided expert-level support and troubleshooting for network-related issues.

- Collaborated with cross-functional teams to design and implement network infrastructure solutions.
- Conducted network assessments and made recommendations for improvements.

USAA via Locus / CDW 4/23 – 12/23

Sr. Network Engineer

Remote

Hired as a contractor of CDW to work on a project for their customer USAA. The project consisted of redesigning the WAN solution and migrating to new Equinix Data Centers utilizing MPLS VPN and IPsec VPN tunnels to connect to USAA business partners. Worked heavily with BGP and OSPF routing protocols, including multiple OSPF areas, AS Prepend and Local Preference and Community Strings to manage and manipulate traffic flow.

- Designed and planned out new network infrastructure, with specific emphasis on layer 2 and layer 3 network technologies
- Designed and implemented upgrade plan for all EoL/EoS network switches on main campus and satellite offices
- Designed and built out a network for new office along with routing for new subnets across all data centers.

AT&T Consulting / Pinnacle 06/22 to 04/23

Lead Network Engineer

Remote

Hired as contractor work for AT&T Consulting Services assigned to several clients and projects.

- Design, architecture, troubleshooting, and support provided to General Dynamics LAN Services.
 - Provided consulting services to design and scope for future growth, configure, support and troubleshoot network, including Palo Alto Firewalls, Cisco ASR and ISR Routers, Cisco Catalyst and Nexus Switches, Cisco FTD Firewalls, Cisco Wireless LAN Controllers, Cisco ISE, BGP, EIGRP, OSPF, MPLS, SDWAN
 - Troubleshooted and Worked with Security issues related to Cisco NGFW FMC and FTD, Checkpoint, Palo Alto firewalls, ASA and Fortigate.
 - Configured AWS Direct Connect and Site-to-Site VPNs for secure, high-throughput hybrid connectivity between on-prem data centers and AWS.
 - Developed Python scripts to automate deployment of standard configurations
- Reclamation, redesign, reconfiguration of network for HP.
 - Analyzed Firewall, Router and Switch configs, Cisco, Palo Alto and Fortinet. Created Low Level Designs and run list configuration for all devices to migrate from public IP scheme to private IP scheme.
 - Performed administrative tasks with Palo Alto Networks (Panorama) including Security, NAT policy

Hagerty / Presidio 09/21 to 06/22

Senior Network Architect

Remote

Hired as a contractor through Presidio to work as a Network Engineer for Hagerty. My primary responsibilities were the design and implementation of Fortinet Firewalls with Fortimanager, Fortinet EMS Client and troubleshooting and resolving various routing and switching issues.

- Designed and documented the architecture and deployment of Fortimanager to manage existing firewalls.
- Designed and documented the architecture and deployment of Forticlient EMS for remote access solutions for remote workers.

- Designed and implemented AWS VPC architectures with subnets, route tables, NAT gateways, security groups, and NACLs for multi-tier application environments.
- Configuration of various network devices including Netscaler, Infoblox, Solar Winds NCM, routers, switches

Bose / Accenture / CDW 06/21 to 09/21

Network Architect

Remote

Hired as a contractor through Accenture to support Bose project, migrating all locations globally from Palo Alto Firewalls to Cisco SDWAN and Umbrella.

- Reviewed existing design and equipment configurations in each location and develop new design and plan to migrate.
- Analyzed existing Palo Alto firewall configs and create policies on Cisco VEdge 100s and 500s using VEdge Cloud Management
- Configured policies and tunnels on Cisco Umbrella
- Configured IPsec tunnels between Cisco Umbrella and VEdge

Salesforce 09/15 – 06/21

Senior Network Engineer

Remote / San Francisco, CA

My primary responsibilities were the architecture, design, and implementation of enterprise IPv6 MPLS, BGP, OSPF backbone consisting of Cisco Nexus 7000 series switches and Cisco ASR 1000 series routers. Configured site to site VPNs between Cisco ASA Firewalls and Juniper SRX Firewalls.

- Designed and implemented Palo Alto (Cloudgenix) SD-WAN solution, migrating network from point-to-point connections and MPLS to an overlay and underlay SDWAN network utilizing IPsec over Public Internet and Private MPLS VPN Network.
- Utilized programming skills to employ Python to automate configuration tasks.
- Migrated legacy on-prem systems to AWS, ensuring network compatibility, security, and compliance with enterprise policies.
- Configured Cisco Identity Services Engine (ISE).
- Configured Cisco Access Control Server (ACS).
- Configured Multi-Protocol BGP in MPLS environment, configuring VRFs and OSPF in a service provider backbone with IP MPLS LDP tag switching.
- Troubleshoot of MPLS, BGP and OSPF routing.
- Configured Cisco Nexus 7010 switches with VPCs and connection to Cisco Nexus Fabric Extenders (FEX).
- Configured Palo Alto PA-400 Series to provide BGP routing and Next Generation Firewall Services between Salesforce IT Network and AWS Direct Connect

CoreDial 07/15 – 09/15

Senior Network Engineer

Blue Bell, PA

Employed by a local regional VoIP Service Provider to assess and optimize the provider's backbone network for reliable, robust, secure, real-time voice traffic. Migrated the network from Cisco Routers and Switches to Juniper MX Routers, EX and QFX Switches. Redesigned the corporate office LAN and Wireless LAN.

- Designed and deployed Wireless LAN with Cisco Wireless LAN Controller 2500 series.
- Converted Cisco Autonomous Access Points to Cisco Lightweight Access Points to work with Controller.
- Configured Juniper SRX Firewalls, MX Routers and EX and QFX Switches.

- Designed a new architecture utilizing new Cisco ASA 5500-X Next-Generation firewalls at every offsite location with its own Internet connection.
- Designed and configured VPN from Cisco ASA to Juniper SRX.
- Managed, supported, optimized SSL VPN platform using Cisco AnyConnect VPN client with various located Cisco ASA firewalls.
- Configured Cisco 6500 switches and Cisco ASR 1000 Switches.
- Cleaned up the data closet by physically removing cables, running new cables, and labeling and documenting all connections in the office data closet.
- Automated network infrastructure provisioning using Terraform and AWS CloudFormation, reducing manual configurations by 75%.

Cisco Systems via GDH Consulting 09/14 - 07/15

Cisco Advanced Services Network Consulting Architect

Remote

Hired as a Red Badge Contractor in the Advanced Services as Network Consulting Engineer to support the Verizon Wireless Account.

- Network design and architecture of migration of Cisco 6500s to Nexus 7710s with Nexus Fabric Extenders.
- Configured and tested Nexus Switches prior to deployment.
- Created configurations and deployed Nexus 7710, Nexus 1000Vs, Nexus Fabric Extenders (FEX) in Verizon Wireless Network.
- Designed and planned Cisco ASA to Cisco Firepower firewall upgrade
- Troubleshoot technical issues and performing root cause analysis.
- Created test plans, implementation plans and methods of procedure (MoP).
- Performed IOS Bug Scrubs.

Synthes / Presidio 03/14 - 09/14

Senior Network Architect

West Chester, PA

Hired by Presidio as a contract consultant to provide network engineering professional services to Synthes during the merger with Johnson and Johnson and integration of two company networks; which includes the design, implementation and support of Cisco based networks; MPLS, Unified Communications, Wireless, Routing, Switching and Security.

- Configured and troubleshoot security policies, VPNs and IPS on Fortinet Fortigate and Cisco ASA Firewalls.
- Configured and troubleshoot Cisco Unified Communications Manager and Unity.
- Configured Fortinet WAF and firewall implementation and configuration for multiple cloud connections
- Configured and troubleshoot Cisco 5508 Wireless LAN Controllers.
- Configured and troubleshoot Cisco Nexus Switches, Cisco Routers and Foundry Switches.
- Configured and troubleshoot Riverbed Steelhead.
- Monitored network using Cisco Prime and WhatsUp Gold.

AT&T / Pinnacle 08/13 - 03/14

Lead Network Architect

Remote

Hired as contractor to be lead engineer on Whirlpool's global network. Design and support of Wireless, MPLS, Routing and Switching.

- Configured authentication, authorization, and accounting (AAA) on Cisco devices using Cisco Secure ACS Radius and TACACS.

- Designed and configured IPv6, MPLS, OSPF/OSPFv3, BGP Backbone for a large enterprise corporation.
- Configured Cisco Wireless LAN Controllers and used troubleshooting skills to resolve technical issues.

Tata Consultancy Services (TCS) / The Fountain Group 10/12 - 08/13

Lead Network Architect

Remote / Cincinnati, OH

Hired as a contract consultant for TCS to work as Lead Architect for GE Aviation. Responsibilities included design, implementation, and support of network projects in the worldwide GE Aviation network.

- Designed networks which included firewalls, wireless devices, switches, routers, load balancers etc. for the deployment and activation of new sites in the GE Aviation Enterprise Network.
- Reviewed and made recommendations of the designs of junior engineers and provided leadership in the design and change approval process.

Cisco Systems / GDH Consulting 12/11 - 10/12

Cisco High Touch Engineer

Philadelphia, PA

Hired as a Red Badge Contractor in the Cisco High-Touch Engineering Team supporting the Sungard account. Responsibilities included network problem resolution, root cause analysis, IOS Software recommendations, architecture, and product training. Act as a focal point for large account network problem resolution and support on a world-wide basis.

- Designed and configured data center solution for customers utilizing Cisco technologies of Overlay Transport Virtualization (OTV), Cisco Fabric Path and Cisco Nexus Virtual Port Channels (VPCs).
- Configured and utilized Cisco Smart Net Total Care Collector to analyze and document network and to document the inventory.
- Performed switch, router, and firewall upgrades to keep with compliance, enable additional features, or as required to meet the needs of project-related tasks
- Provided onsite troubleshooting for Cisco TAC as an onsite High Touch Engineer.
- Resolved complex Cisco TAC cases for various technologies; MPLS, STP, OSPF, EIGRP, BGP, Nexus VPC, 6509 VSS, Cisco Fabric Path and Cisco Overlay Transport Virtualization (OTV).

Amerigas / UGI via TalonPro 05/11 - 12/11

Lead Network Architect

King of Prussia, PA

Hired as network consultant to architect, design, implement and support a network upgrade project involving Data center consolidation and virtualization. Configured Nexus 5510s and 5548s, Nexus 1000Vs, 6509 VSS. Troubleshoot and resolved EIGRP and Spanning Tree issues in a large enterprise network environment. Set up Cisco ACS for TACACS+ and RADIUS Authentication, Authorization and Accounting (AAA) LDAP integration.

- Designed and implemented data center architecture utilizing Cisco Nexus 5596 switches and Cisco Nexus FEXs connected to HP Server Chassis.
- Designed and configured Cisco ACS for all network devices.
- Configured VMWare VSwitch and converted to Cisco Nexus 1000V network.

CPS Energy / Elite Technical and INX Inc. 12/10 - 05/11

Senior Network Security Engineer

San Antonio, TX

Passed the background check and was approved by The North American Electric Reliability Corporation (NERC) Critical Infrastructure Protection (CIP) and hired as a contractor to work onsite at the headquarters of CPS Energy in San Antonio, TX. Duties included design, configuration and support of large Cisco based network used to support a power generation management system. All work performed complied with The North American Electric Reliability Corporation (NERC) Critical Infrastructure Protection (CIP).

- Designed and configured multiple segments of a network that were isolated with high availability Cisco ASA firewalls in active/standby mode.

Macquarie Group / e-Brilliance 06/10 - 12/10

Senior Network Architect

Philadelphia, PA

Hired as a contractor of e-Brilliance to be the lead engineer in the Philadelphia office after Macquarie's acquisition of Delaware Investments and built a new voice and data network and then integrated that network into Macquarie's global network. This integration involved the configuration of Cisco 4900 and 6500 series switches, Cisco 2800 and 3800 routers and voice gateway and Cisco ASA 5510s.

- Configured Cisco routers, switches, ASA Firewalls, Wireless LAN Controllers in a large enterprise network.
- Complied with a very precise and highly detailed and documented change control process.

Pennsylvania, Power and Light (PPL) / Beaumont Staffing 09/09 - 06/10

Senior Network Architect

Allentown, PA

Approved by The North American Electric Reliability Corporation (NERC) Critical Infrastructure Protection (CIP) and hired as a contractor for this large power generation utility corporation to design, implement and support a large enterprise network which included the following protocols and hardware; OSPF, BGP, MPLS, Frame-Relay, ATM and various Cisco routers and switches. Accomplishments included migration of a large Nortel based network to a Cisco based network, converted configurations on Nortel switches and routers and then designed, implemented, and configured a Cisco based network running OSPF, BGP and MPLS.

- Designed and Implemented Wide Area Network Solution by converting spoke sites from a frame-relay connection to an MPLS connection to the hub.
- Migrated large enterprise network from Nortel/Bay/Wellfleet routers and switches to Cisco routers and Cisco Nexus Switches.
- Designed and deployed a large Cisco Multipoint VPN network to connect remote power generation facilities to the core network.

AT&T Consulting / Modis 12/07 - 09/09

Senior Deployment Engineer

Atlanta, GA

Hired as a contractor of AT&T Consulting via Modis to be lead implementation engineer on a very large MPLS implementation. My responsibilities included design verification, testing, installation and configuration of Cisco 12410 GSRs, 7600, 3600s, 2600, 3800s, 2800s, 1800s series routers and 6500s, 4500s, 3500s, 2900s series switches, troubleshooting OSPF, BGP, Multi-protocol BGP and MPLS.

- Configured and implemented the connection of MPLS Provider Edge routers BGP connection to the Customer Edge Routers and configured the MPLS VPN in the AT&T Backbone for a State government network.

Shire Pharmaceuticals / TopGallant 09/06 - 12/07

Cisco IP Telephony Architect

Wayne, PA

Hired by Shire Pharmaceuticals as a consultant to support and maintain a large global Cisco Unified Communications Network.

- Designed and Implemented large Cisco Unified Communications network with Cisco IP Contact Center Express.
- Configured Cisco Unity call flow for Interactive Voice Response and integration with Cisco IP Contact Center.

InfoNXX / KGB USA

IP Telephony Engineer 09/05 – 09/06

Bethlehem, PA

- Configured proprietary SIP phone systems integrated with Cisco ICM for routing calls between multiple call centers that provided director assistance and other information for cell phones dialing 411.

Cisco Systems

Systems Engineer 05/04 - 09/05

Malvern, PA; Edison, NJ

- Provided pre sales support and post sales support to Cisco Resellers and Integrators in the areas of routing, switching and IP Telephony/Voice over IP Solutions.
- Supported account managers on the Comcast account specifically Comcast Internal IT and Comcast Business Communications in all technologies, which included Storage Area Networking, Wireless LANs, Security, IP Telephony, Routing and Switching.

Verizon Enterprise Solutions

Network Consultant Contractor 11/03 – 05/04

Marcus Hook, PA

- Provided Managed Services Consulting as a Verizon Contractor to Sunoco

Verizon Enterprise Solutions

Senior Technology Solutions Architect 09/01 - 11/03

Conshohocken, PA

- Provided pre sales support and post sales support to Cisco Resellers and Integrators in the areas of Routing, Switching and IP Telephony/Voice over IP Solutions.
- Supported all the account managers in PA/DE region in the areas of Wireless Networking, IP Telephony, VoIP, and complex routing and switching. Achieved or exceeded target for two consecutive years.

TruePosition

Senior Design Architect 06/00 - 09/01

King of Prussia, PA

- Designed, configured, and maintained data network for transport of time sensitive data in TruePosition's Wireless Location Systems. Integrated customer networks into the TruePosition Wireless Location System network. Interfaced with the development team to maximize the performance of and document the TruePosition Wireless Location System. Worked with software
- engineers to maximize performance of TruePosition's proprietary applications through the network. Used Sniffer captures extensively to troubleshoot and optimize the network.

Cisco Network Academy, Montgomery County Community College

CCNA Instructor (Part Time) 09/00 - 12/00

Blue Bell, PA

- Taught community college evening course to students on how to install and configure Cisco switches and routers in multiprotocol networks using local and wide-area networks (LANs and WANs).

Suburban Cable/Comcast Cable

Engineering Manager, Cable Modem Termination Systems 11/99 - 06/00

Oaks, PA

- Coordinated a team of network engineers in the installation and configuration of cable modem termination systems throughout a suburban Philadelphia.

AT&T Wireless

Field Operations Manager 03/97 - 11/99

Bensalem, PA

- Directed a team of 15 field engineers supporting, integrating, and maintaining 500+ radio-base stations throughout the Philadelphia metropolitan service area. Project managed the building of new radio-base stations and the modification and replacement of existing radio-base stations. Configured radios, T-1 carrier equipment, multiplexers, and large central office switches.

Education

Associate degree Electronics Engineering Technology
Pennsylvania Institute of Technology

Interests

Private Pilot Single and Multi-Engine
Aircraft Owners and Pilots Association (AOPA)
Tuskegee Airmen Inc.
Health and Fitness
Civil Air Patrol